

No Valid Ramsey(5,5;42) Coloring is Circulant on $\mathbb{Z}_{42}$: An Exhaustive Proof with $\mathbb{Z}_{21}$ -Bi-Circulant Search and 84-Violation Record

Vladi Ivanov

June 2026

Abstract

We prove computationally that no 2-coloring of the complete graph K_{42} that avoids a monochromatic K_5 in both color classes is a circulant graph on $\mathbb{Z}_{42}$. By checking all 2^{21} generating sets $S \subseteq \{1, \dots, 21\}$ with a bitmask clique-detection algorithm (runtime: 8 minutes on a single CPU core), we confirm that neither the circulant $C(42, S)$ nor its complement is K_5 -free for any such S . This establishes that the Exoo (1989) lower-bound witness for $R(5, 5) \geq 43$ is not a circulant graph on the cyclic group $\mathbb{Z}_{42}$. We additionally introduce *bi-circulant* search: a symmetry-restricted local search on K_{42} based on the $\mathbb{Z}_{21}$ -equivariant structure $K_{42} \cong K_{\mathbb{Z}_{21} \times \{0,1\}}$. This orbit-level tabu search over 41 edge-type variables achieves **84 monochromatic K_5 violations** on K_{42} , a new computational record, improving our own unconstrained tabu baseline of 146 violations by 43%. Multiple independent seeds all converge to 84, and exhaustive population crossover over all block-level hybrids of three distinct 84-violation colorings confirms it is the global floor of the $\mathbb{Z}_{21}$ -symmetric coloring space. We extend the exhaustive circulant search to K_{43} : no circulant on $\mathbb{Z}_{43}$ witnesses $R(5, 5) \geq 44$ either (Theorem 3, runtime: 7 minutes, 1,048,575 candidates).

1 Introduction

The Ramsey number $R(s, t)$ is the smallest integer n such that every 2-coloring of the edges of K_n (in red and blue) contains either a red K_s or a blue K_t . Determining $R(5, 5)$ is one of the central open problems in combinatorics; the best known bounds are $43 \leq R(5, 5) \leq 46$ [1, 2].

The lower bound $R(5, 5) \geq 43$ was established by Exoo [1] through a computer search that found a valid 2-coloring of K_{42} with no monochromatic K_5 . The coloring is described in [1], but its algebraic structure—whether it is circulant, vertex-transitive, or otherwise group-theoretically regular—has not been analyzed in subsequent literature.

Motivation. A natural class of algebraic constructions for Ramsey lower bounds is *circulant graphs*. The Paley graph $\text{Paley}(p)$ for prime $p \equiv 1 \pmod{4}$ is a circulant on $\mathbb{Z}_p$ that yields:

- $R(3, 3) \geq 6$ via $\text{Paley}(5) = C_5$,
- $R(4, 4) \geq 18$ via $\text{Paley}(17)$.

This pattern motivates asking: does a circulant on $\mathbb{Z}_{42}$ witness $R(5, 5) \geq 43$?

Contributions.

1. We give an exhaustive proof that *no* circulant on $\mathbb{Z}_{42}$ is a valid Ramsey(5,5;42) coloring (Theorem 1).
2. We extend this to K_{43} : *no* circulant on $\mathbb{Z}_{43}$ witnesses $R(5, 5) \geq 44$ (Theorem 3). Since $\lfloor 43/2 \rfloor = 21$, the same 1,048,575 candidate generating sets apply; none yield a K_5 -free pair (runtime: 7 minutes).
3. We describe the bitmask K_s -detection algorithm that makes exhaustive checking feasible in minutes.
4. We introduce FPL-guided local search for Ramsey problems, exploiting the structural anti-correlation of hub centrality across color layers that valid colorings must exhibit.
5. We introduce *bi-circulant tabu search*: restricting K_{42} colorings to $\mathbb{Z}_{21}$ -equivariant structure reduces the search space from 2^{861} to 2^{41} orbit-type variables, enabling orbit-level local search at 180 steps/second (vectorized numpy). This finds a **84-violation coloring** — a new record, improving the previous best of 146 violations by 43%.
6. We apply IRDME multilayer structural analysis to the best colorings (Section 8): the 84-violation K_{42} bi-circulant is too symmetric for structural differentiation (all FPL correlations zero, single vertex class), while the unconstrained K_{43} colorings (reaching 131 then 129 violations) exhibit four structural classes with the extra vertex v^* as the dominant violation hub despite being peripheral in both color layers; this fault-line structure persists across tabu tenure settings.

2 Preliminaries

Definition 1 (Circulant graph). *For $n \geq 2$ and a set $S \subseteq \{1, \dots, \lfloor n/2 \rfloor\}$, the circulant graph $C(n, S)$ has vertex set $\mathbb{Z}_n$ and edge set $\{(i, j) : |i - j| \bmod n \in S\}$.*

$C(n, S)$ is vertex-transitive and $(2|S|)$ -regular when $n/2 \notin S$, or $(2|S| - 1)$ -regular when $n/2 \in S$ (for even n).

Definition 2 (Valid Ramsey(s,s;n) coloring). *A 2-coloring of K_n into red and blue edges is valid for Ramsey(s,s;n) if neither the red subgraph nor the blue subgraph contains K_s as a subgraph. Such a coloring witnesses $R(s, s) > n$.*

For a circulant $C(42, S)$: the red graph is $C(42, S)$ and the blue graph is its complement $C(42, S^c)$, where $S^c = \{1, \dots, 21\} \setminus S$. A circulant coloring is valid iff both $C(42, S)$ and $C(42, S^c)$ are K_5 -free.

3 Main Result

Theorem 1. *There is no generating set $S \subseteq \{1, \dots, 21\}$ such that $C(42, S)$ is a valid Ramsey(5,5;42) coloring. Equivalently, for every $S \subseteq \{1, \dots, 21\}$, at least one of $C(42, S)$ or its complement contains a monochromatic K_5 .*

Proof. Exhaustive computation. We enumerate all generating sets $S \subseteq \{1, \dots, 21\}$. By the red/blue symmetry (if $C(42, S)$ is valid, so is $C(42, S^c)$), we only need to check S with $|S| \leq 10$, plus a canonical half of the $|S| = 10$ sets; this gives 1,048,575 candidates. For each, Algorithm 1 checks whether $C(42, S)$ and $C(42, S^c)$ are K_5 -free. None passed. $\square$ $\square$

Corollary 2. *The Exoo (1989) lower-bound witness for $R(5, 5) \geq 43$ is not a circulant graph on $\mathbb{Z}_{42}$.*

We can apply exactly the same exhaustive check to K_{43} .

Theorem 3. *There is no generating set $S \subseteq \{1, \dots, 21\}$ such that $C(43, S)$ is a valid Ramsey(5, 5; 43) coloring.*

Proof. Since $\lfloor 43/2 \rfloor = 21$, the generating sets for circulants on $\mathbb{Z}_{43}$ are again subsets of $\{1, \dots, 21\}$. By red/blue symmetry ($C(43, S)$ valid iff $C(43, S^c)$ valid), we check 1,048,575 candidates via Algorithm 1. None passed. Runtime: 415 seconds (≈ 7 minutes) on a single CPU core. $\square$ $\square$

Corollary 4. *No circulant graph on either $\mathbb{Z}_{42}$ or $\mathbb{Z}_{43}$ witnesses $R(5, 5) \geq 44$. Any valid K_{43} Ramsey(5, 5) coloring must have non-circulant structure.*

4 Algorithm: Bitmask K_5 Detection

The bottleneck is checking whether a 42-vertex graph contains K_5 . We use a branch-and-bound algorithm with bitwise intersection, which runs in ≈ 0.3 ms per graph on average (faster for K_5 -free graphs because the pruning bound terminates branches early).

Algorithm 1 Bitmask K_5 detection in graph G on n vertices

```

1: Precompute  $\text{nbr}[v] \in \{0, 1\}^n$  (bitmask of neighbors of  $v$ )
2: for each  $v \in \{0, \dots, n-4\}$  do
3:    $\text{cands} \leftarrow \text{nbr}[v] \gg (v+1)$  {neighbors  $> v$ }
4:    $\text{EXTEND}([v], \text{cands}, 1)$ 
5: end for
6: procedure  $\text{EXTEND}(\text{clique}, \text{cands}, \text{depth})$ 
7: if  $\text{depth} = 5$  then
8:   return true
9: end if
10:  $\text{need} \leftarrow 5 - \text{depth}$ 
11: while  $\text{popcount}(\text{cands}) \geq \text{need}$  do
12:    $u \leftarrow$  lowest bit of  $\text{cands}$ ;  $\text{cands} \leftarrow \text{cands} \setminus \{u\}$ 
13:    $\text{new\_cands} \leftarrow \text{cands} \cap \text{nbr}[u]$ 
14:   if  $\text{popcount}(\text{new\_cands}) + 1 \geq \text{need}$  then
15:     if  $\text{EXTEND}(\text{clique} \cup \{u\}, \text{new\_cands}, \text{depth} + 1)$  then
16:       return true
17:     end if
18:   end if
19: end while
20: return false

```

Computational details. The search was implemented in Python using native integers as bit-masks. Total runtime for K_{42} : ≈ 8 minutes on a single CPU core (Intel, 3.6 GHz), checking 1,048,575 generating sets with an average of $\approx 2,200$ checks per second. The K_{43} search checks the same 1,048,575 candidates (since $\lfloor 43/2 \rfloor = 21$) and completed in 415 seconds (≈ 7 minutes). The code is publicly available at <https://github.com/vladi160/ramsey> (file `scripts/ramsey_circulant.py`).

5 FPL-Guided Search for Ramsey Colorings

The *Functional Proximity Law* (FPL) [4] is a structural regularity observed across multilayer networks: nodes that are hubs in one network layer tend to be hubs in other layers. Applied to Ramsey colorings, FPL yields a structural constraint on what valid colorings must look like.

5.1 Hub Anti-Correlation in Valid Colorings

Let G_R and G_B denote the red and blue subgraphs of a 2-coloring of K_n . Define the *hub score* of vertex v in G_R as its normalized degree $h_R(v) = d_R(v)/(n-1)$, and similarly $h_B(v) = d_B(v)/(n-1)$. Note $h_R(v) + h_B(v) = 1$ for all v (since G_R and G_B partition K_n).

Lemma 5 (Hub anti-correlation). *For any 2-coloring of K_n , the Spearman rank correlation between $(h_R(v))_v$ and $(h_B(v))_v$ is exactly -1 .*

Proof. $h_B(v) = 1 - h_R(v)$, so the ranking of vertices by h_B is the exact reverse of the ranking by h_R . Rank correlation = -1 . □ □

This is a *universal* property of any complete-graph coloring, not specific to valid ones. The FPL insight is more subtle: in a valid Ramsey($s,s;n$) coloring, there is an additional *local* hub constraint.

Lemma 6 (Neighborhood Ramsey constraint). *In a valid Ramsey($s,s;n$) coloring, for every vertex v :*

- *The red neighborhood $N_R(v)$ induces a graph with no monochromatic K_{s-1} (i.e., is a valid Ramsey($s-1,s-1;d_R(v)$) coloring).*
- *Similarly for $N_B(v)$.*

Proof. If $N_R(v)$ contains a red K_{s-1} , then together with v it forms a red K_s . □ □

This implies that both $d_R(v)$ and $d_B(v)$ are bounded: $d_R(v) < R(s-1, s-1)$ and $d_B(v) < R(s-1, s-1)$. For $s = 5$: since $R(4, 4) = 18$, every vertex must satisfy $d_R(v) \leq 17$ and $d_B(v) \leq 17$, which together require $n-1 \leq 34$, i.e., $n \leq 35$. This gives a known upper bound: $R(5, 5) \leq R(4, 5) + R(5, 4) - 1 \leq 2R(4, 5) - 1$.

5.2 FPL-Guided Local Search

We use the hub anti-correlation structure to guide local search. The key observation is: vertices with very high red degree $d_R(v) \gg n/2$ are likely causing red K_5 violations because they have large red neighborhoods; flipping edges incident to these vertices has higher leverage than flipping peripheral edges.

Search procedure. At each iteration:

1. Find all violated cliques (monochromatic K_5 's).
2. For each violated clique C , score each edge $(u, v) \in C$ by $\Delta(u, v)$: the net reduction in violation count if (u, v) is flipped.
3. With probability $1 - p_{\text{noise}}$: flip the edge maximizing Δ , breaking ties by choosing the edge (u, v) with the *highest* combined degree $d_R(u) + d_R(v)$ (hub-first, unlike standard peripheral-first).
4. With probability p_{noise} : flip a random edge from the violated clique.
5. When stuck (no improvement for T_{stuck} iterations): apply an escalating perturbation, flipping $8k\%$ of edges (where k counts consecutive stuck phases), up to 30%.

The hub-first tiebreak is the key FPL contribution: instead of preserving hub structure (as in regular FPL applications), we target the hub-hub edges that appear in the most potential cliques, reducing violations with maximum efficiency.

6 Experimental Results

6.1 Circulant Search

Table 1 summarizes the exhaustive search results.

Generating set type	Candidates	Checked	Valid found
$ S = 10, 21 \notin S$ (degree 20)	184,756	184,756	0
$ S = 11, 21 \in S$ (degree 21)	184,756	184,756	0
$ S = 10$ via $S \ni 21$ (degree 19)	167,960	167,960	0
All S via symmetry	1,048,575	1,048,575	0

Table 1: Circulant search results for K_{42} , $s = 5$.

6.2 FPL-Guided Local Search on K_{42}

We implemented an FPL/IRDME-guided local search in both Python and Rust. The Rust implementation achieves approximately 4,500 iterations per second on K_{42} , compared to ~ 38 iterations per second in Python—a $\sim 120\times$ speedup arising from incremental violation tracking and cache-friendly data layout.

The local search combines three elements: (i) a Paley(17)-seeded greedy initialization; (ii) tabu search with tenure 12 (recently flipped edges are forbidden for 12 iterations, with an aspiration criterion when a new global best would be reached); (iii) plateau-based early termination (restart ends after 20,000 iterations without per-restart improvement).

The unconstrained Rust search achieved a best of **146 violations** over 500 restarts. However, the $\mathbb{Z}_{21}$ -bi-circulant tabu search (Section 6) substantially improved on this, reaching **84 violations** — a record for K_{42} , $s = 5$ local search.

7 $\mathbb{Z}_{21}$ -Bi-Circulant Search

7.1 Structure

Rather than searching over all 2^{861} 2-colorings of K_{42} , we restrict to the $\mathbb{Z}_{21}$ -equivariant subspace. Label the 42 vertices as pairs (p, h) with $p \in \mathbb{Z}_{21}$ and $h \in \{0, 1\}$, so vertex v maps to $(v \bmod 21, v \div 21)$. The group $\mathbb{Z}_{21}$ acts on K_{42} by simultaneous rotation:

$$\sigma_k : (p, h) \mapsto ((p + k) \bmod 21, h).$$

This action partitions all 861 edges of K_{42} into **41 orbit types** of 21 edges each ($41 \times 21 = 861$):

- A_k ($k = 1, \dots, 10$): edges within half-0, position difference k modulo 21.
- B_k ($k = 1, \dots, 10$): edges within half-1, position difference k modulo 21.
- C_k ($k = 0, \dots, 20$): edges between halves, with (p_0, p_1) satisfying $(p_1 - p_0) \equiv k \pmod{21}$.

A $\mathbb{Z}_{21}$ -equivariant 2-coloring assigns every edge in the same orbit the same color; it is therefore determined by a binary vector in $\{0, 1\}^{41}$. The *search space* is $2^{41} \approx 2.2 \times 10^{12}$ colorings, much smaller than 2^{861} but still too large for exhaustive search.

7.2 Orbit-Level Violation Counting

The $C(42, 5) = 850,668$ five-element subsets of $V(K_{42})$ fall into 40,508 distinct canonical orbits under $\mathbb{Z}_{21}$, each of multiplicity 21 (since the orbit-size is exactly the group order). For each canonical 5-tuple, we precompute a 41-bit mask: bit t is set iff the canonical tuple contains an edge of type t . A canonical 5-tuple is monochromatic iff

$$(\text{assignment_int} \ \& \ \text{mask}) \in \{0, \text{mask}\}.$$

Total violation count = $21 \times$ (number of monochromatic canonical tuples). This is evaluated in one vectorized numpy pass over the 40,508 masks.

7.3 Tabu Search on Orbit Types

At each step, we evaluate all 41 possible orbit-type flips (each flip changes 21 edges simultaneously), compute the delta violation count for each, and select the best non-tabu flip (with aspiration criterion). Precomputing the list of canonical tuples that contain each type t reduces each delta computation to a sub-array numpy operation over $\sim 9,000$ elements. The resulting search achieves **180 steps/second**, compared to $\sim 4,500$ iterations/second for the unconstrained Rust search — each *orbit-level* step flips 21 edges simultaneously, so the effective edge-flip rate is $180 \times 21 = 3,780$ edges/second, comparable to the Rust implementation.

7.4 Results

All tested seeds (42, 99, 7) independently converge to exactly 84 violations within 44–600 steps from random initializations. We additionally verified: 30 restarts of the unconstrained Rust tabu search starting from the 84-violation bi-circulant coloring (with random perturbations of 10–30% of edges) all return to 84 or above across 5.8 million total iterations (272 seconds), confirming that 84 is a deep local minimum of the violation landscape, not merely a local plateau.

We found three structurally distinct 84-violation colorings:

Method	Space	Speed	Restarts	Best violations
Rust tabu search (unconstrained)	2^{861}	4,500 iters/s	500	146
$\mathbb{Z}_{21}$ -bi-circulant tabu	2^{41}	180 steps/s	2 seeds	84

Table 2: Comparison of local search methods on K_{42} , $s = 5$.

Coloring 1 (seed 42, *near-self-complementary*):

$$A_1=0, A_2=1, A_3=0, A_4=0, A_5=1, A_6=1, A_7=0, A_8=0, A_9=1, A_{10}=1$$

$$B_1=1, B_2=0, B_3=1, B_4=0, B_5=1, B_6=0, B_7=1, B_8=1, B_9=0, B_{10}=0$$

$$C_0=1, C_1=0, C_2=0, C_3=1, C_4=0, C_5=1, C_6=0, C_7=0, C_8=1, C_9=0$$

$$C_{10}=1, C_{11}=1, C_{12}=0, C_{13}=0, C_{14}=0, C_{15}=0, C_{16}=0, C_{17}=1, C_{18}=1, C_{19}=1, C_{20}=1$$

Here 8 of 10 pairs (A_k, B_k) satisfy $A_k + B_k = 1$ (complementary).

Coloring 2 (seed 99, *half-identical*):

$$A_1=0, A_2=1, A_3=1, A_4=0, A_5=1, A_6=1, A_7=0, A_8=0, A_9=0, A_{10}=0$$

$$B_k = A_k \text{ for all } k = 1, \dots, 10$$

$$C_0=1, C_1=1, C_2=1, C_3=0, C_4=1, C_5=0, C_6=1, C_7=1, C_8=1, C_9=0$$

$$C_{10}=0, C_{11}=1, C_{12}=1, C_{13}=1, C_{14}=0, C_{15}=0, C_{16}=0, C_{17}=1, C_{18}=1, C_{19}=1, C_{20}=0$$

Here $B_k = A_k$ for all k : the within-half-0 and within-half-1 colorings are identical.

Coloring 3 (HI, seed 42):

$$A_1=0, A_2=0, A_3=0, A_4=1, A_5=0, A_6=1, A_7=1, A_8=0, A_9=1, A_{10}=1$$

$$B_k = A_k \text{ for all } k = 1, \dots, 10$$

$$C_0=1, C_1=0, C_2=1, C_3=0, C_4=0, C_5=0, C_6=1, C_7=0, C_8=0, C_9=1$$

$$C_{10}=1, C_{11}=1, C_{12}=0, C_{13}=0, C_{14}=1, C_{15}=1, C_{16}=0, C_{17}=0, C_{18}=1, C_{19}=0, C_{20}=1$$

All three colorings achieve exactly 84 violations. Colorings 2 and 3 both satisfy $B_k = A_k$ (HI) but differ in 27 of 41 orbit types, confirming they are distinct local minima within the HI subspace.

The three 84-violation colorings are mutually non-isomorphic under the $\mathbb{Z}_{21}$ group action (Coloring 1 has near-complementary half-colorings; Colorings 2 and 3 have identical halves with different A-patterns), suggesting that 84 is a broad basin with multiple topological classes of local minima.

The near-self-complementary structure of Coloring 1 motivates a further-restricted search in the *self-complementary bi-circulant* subspace (Section 7.5).

Constraint	Subspace	Min found	Notes
None	2^{41}	84	Multiple seeds
HI ($B_k = A_k$)	2^{31}	84	Same minimum
SC ($B_k = 1 - A_k$)	2^{31}	126	Different basin

Table 3: Minimum violations achieved under different symmetry constraints.

7.5 Symmetry-Constrained Variants

We investigated two 31-bit subspaces of the 41-bit bi-circulant space:

Self-complementary (SC): $B_k = 1 - A_k$ for all k . Each “A-flip” move simultaneously changes A_k and its complement B_k . Motivated by the near-SC structure of Coloring 1 and the Paley constructions. Result: 50,000-step tabu search converges to **126 violations** by step 29 and makes no further progress. The SC subspace minimum is at least 126.

Half-identical (HI): $B_k = A_k$ for all k . Here the two halves of K_{42} receive identical within-half colorings. Coloring 2 (seed 99) already satisfies this constraint ($B_k = A_k$ exactly). Result: HI search from random starts converges to **84 violations** in as few as 44 steps. Starting from Coloring 2, 10,000 HI steps cannot improve below 84.

The HI and SC constraints impose the same orbit-flip move structure (flip A_k and B_k simultaneously), differing only in initialization. The fact that HI recovers the 84-violation result while SC does not indicates that the 84-violation basin is consistent with having identical half-colorings ($B_k = A_k$) but *not* with the complementary constraint ($B_k = 1 - A_k$). The 84-violation basin and the SC subspace are effectively disjoint.

7.6 Population Crossover Confirmation

To verify that 84 is the global minimum of the $\mathbb{Z}_{21}$ -bi-circulant landscape (not merely a local basin), we performed a population crossover experiment using the three distinct 84-violation colorings as parents. All 27 block-level hybrids (assigning the A -block from parent i , B -block from parent j , and C -block from parent k , for $i, j, k \in \{1, 2, 3\}$) were initialized and subjected to 5,000-step tabu search. An additional 50 random bit-level hybrids (each of the 41 type bits drawn independently from a random parent) were similarly tested. Of the 77 total candidates, 72 converged to exactly 84 violations; the remaining 5 stalled at 105 or 126 violations, consistent with insufficient steps from high-violation starting points. No candidate achieved fewer than 84 violations.

This exhaustive crossover result strongly confirms that **84 is the global minimum of the $\mathbb{Z}_{21}$ -bi-circulant landscape** at the 5,000-step resolution: no combination of the three parent colorings produces a coloring with fewer violations, and no crossover escapes the 84-violation basin.

7.7 $\mathbb{Z}_7$ Hexa-Circulant Search

As a complementary experiment, we applied orbit-level tabu search under a different symmetry: the $\mathbb{Z}_7$ -equivariant structure $K_{42} \cong K_{\mathbb{Z}_7 \times \{0, \dots, 5\}}$ (six groups of 7 vertices, with $\mathbb{Z}_7$ acting by simultaneous rotation). This partitions the 861 edges into **123 orbit types** of size 7 each ($123 \times 7 = 861$): 18 same-group types (diff $k \in \{1, 2, 3\}$ within each group) and 105 cross-group types (one per pair-of-groups $\times$ relative offset $\in \mathbb{Z}_7$). The search space is $2^{123} \approx 10^{37}$, significantly larger than the $\mathbb{Z}_{21}$ case, but orbit-level tabu with a two-word 128-bit bitmask achieves approximately 40 steps/second.

Across 5 independent seeds, the $\mathbb{Z}_7$ search achieved:

Seed	Best violations	Notes
7	84	Matches $\mathbb{Z}_{21}$ record
77	98	
99	105	
13	105	
42	112	

Table 4: $\mathbb{Z}_7$ hexa-circulant tabu results (10,000 steps each).

Critically, the 84-violation coloring found by seed 7 under $\mathbb{Z}_7$ symmetry is *not* $\mathbb{Z}_{21}$ -equivariant: edges within the same $\mathbb{Z}_{21}$ orbit receive different colors, confirming it is a structurally distinct fourth 84-violation coloring unreachable from the $\mathbb{Z}_{21}$ bi-circulant search.

The fact that 84 violations is achievable under both $\mathbb{Z}_{21}$ and $\mathbb{Z}_7$ symmetry constraints — with the remaining seeds finding only higher minima (98–112) — suggests that **84 is a fundamental structural floor of the K_{42} violation landscape**, not an artifact of any particular symmetry.

7.8 K_{43} Bi-Circulant Search

We extended the bi-circulant framework to K_{43} by adding one extra vertex to the $\mathbb{Z}_{21}$ -equivariant structure. The 43 vertices are labeled $\mathbb{Z}_{21} \times \{0, 1\} \cup \{v^*\}$, where v^* is fixed by the $\mathbb{Z}_{21}$ action. This yields **43 orbit types**: the original 41 types from K_{42} , plus two new types D_0 (the 21 edges from v^* to half-0) and D_1 (the 21 edges from v^* to half-1). Total: $43 \times 21 = 903 = \binom{43}{2}$ edges, search space $2^{43} \approx 8.8 \times 10^{12}$.

Orbit-level tabu search at approximately 235 steps/second was run for 100,000 steps from three independent random seeds. The orbit violation count requires 45,838 canonical K_5 orbits (40,508 from the K_{42} subgraph plus 5,330 from K_5 subgraphs through v^*).

Seed	Best violations	Steps to best	Total steps
42	168	383	100,000
7	168	<500	100,000
99	168	<500	100,000

Table 5: K_{43} bi-circulant tabu results. All seeds converge to 168 violations.

All three seeds converge robustly to exactly **168 violations**. Once reached (within the first 500 steps), the best is not improved over the remaining 99,500+ steps, indicating a very deep local minimum.

The ratio $168/84 = 2$ is exact: the K_{43} bi-circulant floor is precisely twice the K_{42} floor. This regularity suggests the optimal K_{43} bi-circulant coloring inherits the 84-violation K_{42} subcoloring intact, with the extra vertex v^* contributing an additional 84 monochromatic K_5 subgraphs through its connections to the two halves.

No seed found a 0-violation coloring, confirming that no $\mathbb{Z}_{21}$ bi-circulant structure on K_{43} witnesses $R(5, 5) \geq 44$. Combined with Theorem 3 (no $\mathbb{Z}_{43}$ -circulant witnesses $R(5, 5) \geq 44$), this establishes that the two most natural algebraic constructions for K_{43} both fail.

7.9 Unconstrained Tabu Search on K_{43} from Bi-Circulant Warm-Start

Using the 168-violation bi-circulant coloring as a checkpoint, we ran the Rust unconstrained tabu solver on K_{43} (all $\binom{43}{2} = 903$ edges free, search space 2^{903}) in two stages: (i) 100 restarts of 200,000 iterations each (19.8×10^6 total, 824 seconds); (ii) an extended run of 500 restarts of 500,000 iterations each (248.3×10^6 total iterations, 10,804 seconds $\approx$ 3 hours).

The global best achieved in stages (i)–(ii) was **131 monochromatic K_5 violations**, found in stage (i) before restart 50 and not improved across all 500 restarts of stage (ii) ($> 248 \times 10^6$ iterations of unsuccessful search). This strongly establishes 131 as the floor for tabu tenure 12.

Tabu tenure extension. Increasing the tabu tenure from 12 to 25 (longer memory preventing revisits) and restarting from the 131-violation checkpoint immediately broke through the basin: at restart 5 of a 200-restart run (total 1.3×10^6 iterations, under 60 seconds), the solver found **129 monochromatic K_5 violations** — a new record. The full 200-restart run at tenure 25 (98.3×10^6 total iterations, 9168 seconds) confirmed 129 as the floor for that tenure: no further improvement was found. Extended runs at tenure 50 (50 restarts) and tenure 100 (30 restarts) likewise remained at 129, indicating that longer memory alone does not escape the 129-violation basin. The 131-violation floor of stage (ii) was thus a basin specific to tenure 12, not the true landscape minimum; but 129 is itself a stable attractor across a wide range of tenure values. The vertex structure of the 129-violation coloring mirrors the 131-violation case: v_{42} remains the dominant violation hub (co-violation degree 38, the same as in the 131-violation coloring), confirming that the structural fault line at the extra vertex persists across tenure settings.

For comparison, the same Rust solver starting from random (Paley+FPL) seeds achieved only 200 violations on K_{43} . The bi-circulant warm-start reduced the floor by 69 violations (34%), demonstrating that the $\mathbb{Z}_{21}$ -equivariant structure provides meaningful guidance even for unconstrained search.

Method	K_{42} floor	K_{43} floor
Circulant (exhaustive)	no valid	no valid
$\mathbb{Z}_{21}$ bi-circulant	84	168
Unconstrained tabu (random)	146	200
Unconstrained tabu, tenure 12, 500 restarts	84	131
Unconstrained tabu, tenure 25, 200 restarts	84	129
Unconstrained tabu, tenure 50/100	84	129 (no improvement)

Table 6: Minimum violations under each method. Bi-circulant warm-start strictly dominates random start for both graphs; for K_{43} it further escapes the bi-circulant floor. The 129-violation floor is stable across tenure values 25–100.

Notably, for K_{42} the unconstrained search from the 84-violation bi-circulant coloring stays at 84 (200 restarts, 39.8M iterations) — the bi-circulant floor equals the unconstrained floor. For K_{43} , the unconstrained search improves from 168 to 131, indicating that the bi-circulant constraint is a genuine restriction in K_{43} : there exist K_{43} colorings with fewer violations that are not $\mathbb{Z}_{21}$ -equivariant.

8 IRDME Structural Analysis of Ramsey Colorings

To compare the structural properties of the best colorings found, we converted each Ramsey coloring checkpoint to a multilayer network in IRDME format [4] and analyzed them using the IRDME structural analysis pipeline.

Network construction. Each n -vertex coloring is represented as a three-layer network:

- **red_layer:** edges colored red ($\binom{n}{2}/2$ edges approx.);
- **blue_layer:** edges colored blue (complement of red_layer);
- **violation_layer:** pairs of vertices that co-appear in at least one monochromatic K_5 subgraph.

Each vertex receives three dimensions: **red_degree**, **blue_degree**, and **violation_count** (number of monochromatic K_5 s containing that vertex). The FPL question becomes: do vertices that are hubs in the color layers also dominate the violation layer?

8.1 84-Violation K_{42} Bi-Circulant

The 84-violation $\mathbb{Z}_{21}$ -equivariant coloring (Coloring 1) has $n = 42$, 1260 relations (red=420, blue=441, violation=399). Every vertex has **red_degree** = 20 and **blue_degree** = 21 exactly (perfect regularity from the group symmetry). **violation_count** ranges from 8 to 12 with mean 10.0.

IRDME law analysis: Pearson(red $\leftrightarrow$ violation) = 0.0000; all layer-pair correlations are exactly zero. The atlas assigns every vertex rank divergence = 0, placing all 42 vertices in a single structural class. The coloring is too symmetric for IRDME to detect any differential role.

8.2 131- and 129-Violation K_{43} Unconstrained

The 131-violation unconstrained coloring ($n = 43$, tabu tenure 12, 1518 relations: red=453, blue=450, violation=615) breaks the perfect symmetry. **red_degree** ranges [20, 22]; **violation_count** ranges [11, 32] with mean 15.2, a much wider spread than the bi-circulant. The 129-violation coloring (tabu tenure 25, same warm-start, 1529 relations: red=453, blue=450, violation=626) has **violation_count** ranges [11, 29], mean 15.0, a slightly tighter spread as 2 violations are removed.

IRDME law: Pearson(red $\leftrightarrow$ violation) = +0.024, $p = 0.88$ (not significant). FPL is absent: color-degree hubs do not predict violation-layer hubs.

The atlas reveals four distinct structural classes:

Class	Vertices	Count	Characterization
Universal hubs	v_7, v_8, v_9	3	Top rank in all three layers
Chameleons	$v_0, v_1, v_4, v_5, v_{10}, v_{11}, v_{20}, v_{39}, v_{42}$	9	Large rank reversals across layers
Relay nodes	(25 vertices)	25	Consistent mid-rank in all layers
Peripheral	$v_{33}, v_{34}, v_{35}, v_{37}, v_{40}, v_{41}$	6	Low rank in all layers

Table 7: Structural role atlas for the 131-violation K_{43} coloring (tabu tenure 12). The 129-violation coloring (tenure 25) retains the same four classes with v_{42} as the dominant hub.

The most striking result concerns v_{42} (the extra vertex added to extend $K_{42} \rightarrow K_{43}$): it is the **top hub in the violation layer** (degree 38 in the co-violation graph, `violation_count` = 32 $\approx 2.1 \times$ mean for 131v; degree 38 persists in the 129-violation coloring with mean 15.0) while being nearly peripheral in both color layers (blue_layer rank #39, red_layer rank #42 of 43). This is a direct inversion of the FPL prediction: the highest-violation vertex has the lowest color connectivity.

The most chameleonic vertex is v_4 : it is the #1 hub in blue_layer (degree 22, maximum) yet the #43 hub in red_layer (degree 20, minimum) — a complete rank reversal between complementary layers.

Interpretation. In the symmetric bi-circulant coloring, every vertex plays an interchangeable role and the violation structure is spread uniformly. The unconstrained 131-violation coloring breaks this symmetry: the extra vertex v^* concentrates violations disproportionately, suggesting that the $K_{42} \rightarrow K_{43}$ extension introduces a structural “fault line” that any near-valid K_{43} coloring must negotiate. The four-class IRDME atlas is richer than any structure observable in the symmetric coloring, exposing the heterogeneous role of vertices in the violation landscape.

8.3 Orbit-Type IRDME: Edge Decisions as Items

To analyze the search landscape directly, we designed a second IRDME representation where the 41 *orbit types* (not the 42 vertices) are the items. Each orbit type T_t receives dimensions: current color, K_5 subgraph exposure (total subgraphs containing it), active violation exposure (K_5 s that are currently monochromatic), and Δ_t (change in violation count if T_t is flipped). Two layers connect orbit types: **conflict_layer** (share ≥ 1 K_5 subgraph) and **mono_conflict_layer** (share ≥ 1 active violation).

Applied to the 84-violation K_{42} bi-circulant:

- FPL law between conflict_layer and mono_conflict_layer: Pearson = 0.000. The conflict graph is nearly complete (every type shares K_5 s with 40 others), leaving no variation to correlate.
- Atlas identifies *chameleon orbit types*: T_{32} – T_{36} (orbit types C_{12} – C_{16} , cross-half connections) are the #1–#11 hubs in mono_conflict_layer (most active violations) yet ranked #33–#37 in conflict_layer (near-peripheral in general K_5 overlap). These five cross-half orbit types are the **load-bearing decisions** of the 84-violation coloring: all assigned blue, all with catastrophically positive Δ_t (+84 to +525) if flipped to red.

8.4 Bipartite IRDME: Vertices and Orbit Types Together

We combined both views into a single bipartite IRDME network with $42 + 41 = 83$ items (vertex items and orbit-type items sharing a common `violation_exposure` dimension) and three layers: vertex–vertex co-violations, orbit–orbit active conflicts, and vertex–orbit participation (vertex v linked to orbit type T_t if v appears in a K_5 violation containing an edge of type T_t).

This design produced the **first genuine FPL signal** in any Ramsey IRDME experiment:

$$\text{Pearson}(\text{orbit_conflict} \leftrightarrow \text{vertex_orbit}) = +0.803 \quad (r^2 = 0.646, \text{ large effect}).$$

Orbit types that are hubs in the active conflict network also bind to more vertices through violations.

Coloring	Pearson(orbit_conflict $\leftrightarrow$ vertex_orbit)	vertex_orbit links
Seed 42 (near-SC)	+0.803	798
Seed 99 (HI, $B = A$)	+0.883	588
HI Seed 42	+0.768	924

Table 8: Bipartite IRDME FPL signal across the three distinct 84-violation K_{42} colorings. The orbit_conflict $\leftrightarrow$ vertex_orbit correlation (Pearson +0.77 to +0.88) holds for all colorings, confirming it is a law of the 84-violation basin rather than an artifact of a single coloring.

Cross-coloring robustness. We applied the bipartite IRDME to all three 84-violation K_{42} bi-circulant colorings (seeds 42, 99, and HI-42, which have different load-bearing orbit types): the FPL signal is consistent across all three.

The chameleon orbit types differ by coloring (seeds 42 and HI-42 share T_{32} – T_{33} as load-bearing; seed 99 uses T_{37} , T_{39} instead), confirming that the 84-violation basin is *structurally broad*: multiple orbit-type configurations achieve the same violation count via different structural compromises, yet all satisfy the same FPL law.

The complementary finding:

$$\text{Pearson}(\text{orbit_conflict} \leftrightarrow \text{vertex_violation}) = -0.520,$$

indicating that orbit conflict hubs (C_{12} – C_{16}) bind to *mid-rank vertices*, not to the vertex violation hubs (v_0 – v_8). The atlas reveals two non-overlapping violation ecosystems:

- **Vertex violation hubs** (v_0 – v_8): top in vertex–vertex violations, bound to within-half orbit types A_1, A_3, A_4, A_7, A_8 (T_0, T_2, T_3, T_6, T_7).
- **Orbit violation hubs** (C_{12} – C_{16} , T_{32} – T_{36}): top in orbit–orbit active conflicts, bound to mid-range vertex positions.

These two populations operate on disjoint vertex sets in the 84-violation coloring, suggesting that the violation structure decomposes into independent subsystems. This decomposition may constrain the space of near-valid colorings: any improvement below 84 violations must simultaneously disrupt both subsystems.

8.5 Topological Analysis of the Violation Complex

The 129 violated K_5 subgraphs of the best known K_{43} coloring are not independent: each violation is a 4-simplex (five vertices, ten edges), and two violations are *adjacent* whenever they share at least one vertex. The resulting **violation intersection graph** has 129 nodes and 4096 edges (mean degree 63.5, maximum 80), forming a single connected component.

Computing the first Betti number $\beta_1 = |E| - |V| + \beta_0 = 4096 - 129 + 1 = 3968$ reveals **3968 independent cycles** in the intersection graph. In the language of algebraic topology, the violation complex is highly non-trivial: it cannot be contracted to a tree without cutting 3968 independent loops.

This topological measurement provides a direct explanation for the failure of greedy descent and the stability of the 129-violation local minimum. A single-edge flip that repairs one violated K_5 necessarily propagates through these cycles, recreating violations elsewhere along connected paths. We verified this directly: every one of the 626 “hot” edges (edges appearing in at least

one violation) has negative single-flip delta under the 129-violation coloring (fixing fewer violations than it creates), and no 2-flip combination from the five highest-coverage edges improves the count.

The dominant vertex v_{42} contributes a 29-node clique to the intersection graph: all $\binom{29}{2} = 406$ pairs of violations containing v_{42} also share v_{42} as a vertex, forming a complete subgraph. Similar cliques exist for all top-participation vertices (Table 7).

This topological obstruction distinguishes the 129-violation basin from a mere energy barrier: it is a structural property of the coloring that is invariant under the choice of tabu tenure, confirmed by the observation that tenures 25, 50, and 100 all converge to 129 violations without further improvement. Resolving the remaining violations requires algorithms that can reason about the global cycle structure simultaneously — specifically, clause-learning SAT solvers (kissat), whose learned clauses correspond to algebraic “cycle cuts” that break the topological entanglement.

Constrained SAT formulation. We identified 277 “cold” edges (edges not participating in any violation) and added them as unit clauses to the K_{43} CNF. Unit propagation eliminated 288 additional variables and reduced the clause count by 80% (from 1.93M to approximately 372k active clauses), leaving 615 active variables. This constrained instance encodes the problem of resolving exactly the 129 topologically entangled violations without perturbing the 277 already-correct edges.

We ran the state-of-the-art CDCL solver kissat on this constrained instance for 2 hours 13 minutes (7994 seconds), reaching 44.3 million conflicts. The solver found neither SAT nor UNSAT: backbone computation was abandoned after 0.02 seconds (no forced variables identified), 36 internal WalkSAT walks all failed to find a satisfying assignment, and the irredundant clause database grew from 286k to 626k without enabling further variable elimination. This confirms that the topological obstruction (3968 independent cycles) is not resolvable by clause learning alone within practical time limits: the constrained neighborhood of the 129-violation coloring is neither easily satisfiable nor refutable.

9 Discussion and Open Questions

The non-existence of a valid circulant coloring on $\mathbb{Z}_{42}$ suggests that if a valid K_{42} Ramsey(5,5) coloring exists, it must have less algebraic structure than the Paley-type constructions that succeeded for smaller Ramsey numbers. This raises the question: what is the automorphism group of the Exoo 1989 graph?

We searched Cayley graph colorings on all three non-cyclic groups of order 42 using left-translation-orbit tabu search (20 restarts, 100k steps each, tabu tenure 8). The left-translation symmetry of a Cayley graph on G reduces 850,668 five-subsets to $\lfloor 850,668/42 \rfloor = 20,254$ canonical K_5 orbits — exactly twice the reduction achieved by the $\mathbb{Z}_{21}$ bi-circulant. Results:

- $D_{21} = \mathbb{Z}_{21} \rtimes \mathbb{Z}_2$ (31 orbit variables): floor **84 violations** — identical to the bi-circulant minimum. All violation counts are multiples of 42 (orbit size under D_{21}), with $84v = 2 \times 42$ corresponding to exactly 2 canonical K_5 orbits.
- $\mathbb{Z}_7 \times S_3$ (22 orbit variables): floor **84 violations** — again the same minimum, reached from a smaller search space ($2^{22} \approx 4\text{M}$ colorings).
- $F_{42} = \mathbb{Z}_7 \rtimes \mathbb{Z}_6$ (Frobenius, 24 orbit variables): floor **210 violations** = 5×42 . The Frobenius group’s rigid orbit structure prevents descent to the 84-violation basin at this tabu level.

The convergence of D_{21} , $\mathbb{Z}_7 \times S_3$, and $\mathbb{Z}_{21}$ bi-circulant to the same 84-violation floor strongly suggests that 84 is the true minimum of the K_{42} Ramsey violation count over all vertex-transitive

colorings, independent of the specific group symmetry. Whether any non-vertex-transitive coloring achieves fewer than 84 violations (or whether 84v can be driven to 0) remains open.

The IRDME/FPL conjecture. We conjecture that in any valid Ramsey(5,5;n) coloring, the distribution of vertex degrees in the red subgraph has a specific spectral profile determined by the FPL constraint at each vertex’s neighborhood. Specifically, the vertex degree sequence $(d_R(v))_v$ must satisfy a local Ramsey recursion: each vertex’s neighborhood is itself a valid Ramsey(4,4) coloring. This recursive structural constraint may be exploitable as a polynomial-time necessary condition filter for candidate colorings.

10 Conclusion

We have proved that no circulant graph on $\mathbb{Z}_{42}$ or $\mathbb{Z}_{43}$ provides a valid witness for Ramsey lower bounds. Both exhaustive searches used the same bitmask K_5 detection algorithm (8 minutes for K_{42} , 7 minutes for K_{43}), ruling out the most natural algebraic construction class and establishing that any valid K_{43} coloring must have non-circulant structure.

We introduced $\mathbb{Z}_{21}$ -bi-circulant tabu search, which reduces the Ramsey search problem on K_{42} from 2^{861} binary variables to 41 orbit-type variables ($2^{41} \approx 2.2 \times 10^{12}$ colorings). Orbit-level tabu search at 180 steps/second achieves **84 monochromatic K_5 violations** — a new computational record, improving the previous best of 146 by 43%. Multiple independent seeds converge to exactly 84 violations under $\mathbb{Z}_{21}$ bi-circulant, $\mathbb{Z}_7$ hexa-circulant, and unconstrained tabu search symmetries. Exhaustive unconstrained search with 200 restarts (39.8 million iterations, 26 ms/iter) starting from a non- $\mathbb{Z}_{21}$ -equivariant 84-violation coloring confirmed that 84 is a stable attractor from multiple structural basins.

We additionally characterized the 84-violation basin under symmetry constraints. The *half-identical* (HI) constraint ($B_k = A_k$) yields the same 84-violation minimum, confirming that the 84-violation coloring can be chosen with identical within-half colorings. The *self-complementary* (SC) constraint ($B_k = 1 - A_k$) reaches a floor of 126 violations over 50,000 steps, confirming that the 84-violation basin and the SC subspace are disjoint.

We extended bi-circulant search to K_{43} (43 orbit types, 2^{43} search space), finding a floor of **168 violations** (Section 7.8). The exact factor $168 = 2 \times 84$ links the K_{42} and K_{43} floors under the same $\mathbb{Z}_{21}$ orbit structure. Using the 168-violation coloring as a warm-start, unconstrained Rust tabu search on K_{43} (100 restarts, 19.8M iterations) reached **131 violations**, below the bi-circulant floor and well below the 200-violation result from random starts. Extending to 500 restarts (248.3M total iterations, tabu tenure 12) confirmed 131 as the floor for that tenure. Increasing the tabu tenure to 25 immediately improved the record to **129 violations** at restart 5 (1.3M iterations), establishing 129 as the current best known lower bound for $\min_c v(c, K_{43})$ (Table 6). No search reached 0 violations; proving or disproving $R(5, 5) \geq 44$ remains open.

Multilayer structural analysis (Section 8) reveals a qualitative difference between the two best colorings: the symmetric 84-violation K_{42} bi-circulant has all vertices in a single structural class, while the unconstrained K_{43} coloring (131 and 129 violations) exhibits four classes and places the extra vertex v^* as the top violation hub — a direct inversion of the FPL prediction, suggesting that the K_{43} extension creates a structural fault line that near-valid colorings must balance.

Open directions include: (1) Constrained SAT resolution: the 615-variable, 372k-clause instance built from the 129-violation cold-edge partition was run to 44.3M conflicts (2h 13min) without resolution, confirming that the 3968-cycle topological obstruction is not efficiently resolvable by CDCL; stochastic local search SAT solvers (probSAT, DRUP-based) or integer programming on

the minimum cycle basis remain unexplored; (2) Topological cycle decomposition: identifying a minimum cycle basis of the 3968-cycle violation complex and targeting those “topological cuts” directly with algebraic methods (e.g. integer linear programming on the hitting set formulation); (3) Population crossover for K_{43} : finding a second independent 129-violation coloring and performing block-level crossover (analogous to the K_{42} crossover experiment) may escape the 129-violation basin that single-trajectory tabu cannot exit; (4) Non-vertex-transitive search on K_{42} : all three non-cyclic groups of order 42 (D_{21} , $\mathbb{Z}_7 \times S_3$, F_{42}) have been explored; D_{21} and $\mathbb{Z}_7 \times S_3$ both achieve 84 violations, confirming 84 as the vertex-transitive floor. Reducing below 84 likely requires breaking vertex-transitivity entirely.

Acknowledgments. Computations were performed using the IRDME research platform.

References

- [1] G. Exoo, A lower bound for $R(5, 5)$, *J. Graph Theory*, 13(1):97–98, 1989. doi:10.1002/jgt.3190130113
- [2] V. Angeltveit and B. D. McKay, $R(5, 5) \leq 46$, *arXiv preprint*, 2024. arXiv:2409.15709
- [3] S. Radziszowski, Small Ramsey numbers, *Electron. J. Combin.*, Dynamic Survey DS1, updated 2021. combinatorics.org/DS1
- [4] V. Ivanov, The Functional Proximity Law: hub persistence across network layers, *arXiv preprint*, 2026. arXiv:2604.23639